

Table S1. Life traits of ant species considered in this study. Information was derived from personal observations, from the literature, and from personal communications from various colleagues. Polymorphism: 1, monomorphism; 2, low polymorphism; 3, high polymorphism. Colony population: 1, hundreds; 2, thousands; 3, tens of thousands. Brood cycle: 1, larvae in the nests from March to September; 2, larvae in the nests during the whole year. Defence of resources: 1, none; 2, food and/or territory. Daily activity rhythms: D, diurnal; DC, diurnal in spring and continuous or with a midday drop in summer; CN, continuous in spring and nocturnal in summer; N, nocturnal. CTM, critical thermal maximum (in °C). MAT, maximum activity temperature (in °C). *: major workers; **: minor workers.

Species	Mean Worker length (mm)	Polymorphism	Colony population	Brood cycle	N queens per colony	N nests per colony	Main food resource
<i>Linepithema humile</i>	2.4	1	3	2	polygyny	polydomy	honeydew
<i>Tapinoma nigerimum</i>	4.0	2	3	2	polygyny	polydomy	honeydew
<i>Camponotus cruentatus</i>	10.0	3	2	2	monogyny	monodomy	honeydew
<i>Camponotus foreli</i>	7.0	3	1	2	monogyny	polydomy	nectar
<i>Camponotus piceus</i>	5.2	2	1	2	monogyny	monodomy	nectar
<i>Camponotus sylvaticus</i>	9.3	3	2	2	monogyny	monodomy	honeydew
<i>Cataglyphis cursor</i>	5.8	2	1	1	monogyny	monodomy	insects
<i>Cataglyphis floricola</i>	6.0	1	1	1	monogyny	monodomy	insects
<i>Cataglyphis iberica</i>	7.6	3	1	1	monogyny	polydomy	insects
<i>Cataglyphis rosenhaueri</i>	5.7	2	1	1	monogyny	polydomy	insects
<i>Cataglyphis velox</i>	8.3	3	1	1	polygyny	polydomy	insects
<i>Formica subrufa</i>	4.9	1	1	1	monogyny	monodomy	insects
<i>Plagiolepis pygmaea</i>	1.6	1	1	2	polygyny	monodomy	nectar
<i>Proformica nasuta</i>	4.5	3	1	1	monogyny	monodomy	insects
<i>Aphaenogaster gibbosa</i>	4.9	1	1	2	monogyny	monodomy	insects
<i>Aphaenogaster senilis</i>	7.1	1	1	2	monogyny	monodomy	insects
<i>Crematogaster auberti</i>	3.4	1	2	2	monogyny	monodomy	nectar
<i>Crematogaster sordidula</i>	2.5	1	1	2	monogyny	monodomy	nectar
<i>Leptothorax kraussei</i>	2.8	1	1	2	monogyny	monodomy	nectar
<i>Leptothorax specularis</i>	2.5	1	1	2	monogyny	monodomy	nectar
<i>Leptothorax unifasciatus</i>	2.5	1	1	2	monogyny	monodomy	nectar
<i>Messor barbarus</i>	7.9	3	2	2	monogyny	monodomy	seeds
<i>Messor bouvieri</i>	6.3	2	2	2	monogyny	monodomy	seeds
<i>Messor capitatus</i>	8.4	3	2	2	monogyny	monodomy	seeds
<i>Messor lusitanicus</i>	6.5	2	2	2	monogyny	monodomy	seeds
<i>Myrmica sabuleti</i>	4.4	1	1	2	polygyny	monodomy	insects
<i>Pheidole pallidula</i>	1.9*/4.1**	3	3	1	monogyny	monodomy	insects
<i>Tetramorium caespitum</i>	2.9	1	2	2	monogyny	monodomy	insects
<i>Tetramorium impurum</i>	3.4	1	2	2	monogyny	monodomy	insects
<i>Tetramorium semilaeve</i>	2.6	1	2	2	polygyny	monodomy	insects

Online resource. Continuation

Species	Defense of resources	Mean Number workers/bait	Daily rhythm	Month of peak activity	CTM (°C)	MAT (°C)	Phylogeny code
<i>Linepithema humile</i>	2	40	CN	September	40	20	BBB
<i>Tapinoma nigerimum</i>	2	43	CN	June	42	24	BBA
<i>Camponotus cruentatus</i>	2	8	DC	July	48	40	BAAAAAB
<i>Camponotus foreli</i>	1	6	D	August	48	44	BAAAAAA
<i>Camponotus piceus</i>	1	4	D	July	44	30	BAAAB
<i>Camponotus sylvaticus</i>	2	9	N	August	46	28	BAAAAAB
<i>Cataglyphis cursor</i>	1	3	D	July	50	48	BAABBBAA
<i>Cataglyphis floricola</i>	1	2	D	July	50	44	BAABBBAB
<i>Cataglyphis iberica</i>	1	3	D	July	52	50	BAABBBBAA
<i>Cataglyphis rosenhaueri</i>	1	1	D	July	50	44	BAABBBBAB
<i>Cataglyphis velox</i>	1	2	D	July	52	46	BAABBBBBB
<i>Formica subrufa</i>	2	7	D	July	48	40	BAABA
<i>Plagiolepis pygmaea</i>	1	9	CN	May	40	22	BAB
<i>Proformica nasuta</i>	1	3	D	June	46	36	BAABBA
<i>Aphaenogaster gibbosa</i>	1	15	D	July	42	36	ABABBB
<i>Aphaenogaster senilis</i>	1	11	D	June	46	42	ABABBA
<i>Crematogaster auberti</i>	2	33	CN	July	44	24	ABBBBA
<i>Crematogaster sordidula</i>	2	24	CN	May	40	20	ABBBB
<i>Leptothorax kraussei</i>	1	2	CN	June	42	26	ABBAC
<i>Leptothorax specularis</i>	1	2	CN	June	42	24	ABBAA
<i>Leptothorax unifasciatus</i>	1	2	CN	June	42	24	ABBAB
<i>Messor barbarus</i>	2	12	DC	September	44	18	ABABAAB
<i>Messor bouvieri</i>	1	26	D	September	44	30	ABABABA
<i>Messor capitatus</i>	2	15	DC	October	44	22	ABABAAA
<i>Messor lusitanicus</i>	1	25	D	September	44	30	ABABABB
<i>Myrmica sabuleti</i>	2	17	N	June	40	22	AA
<i>Pheidole pallidula</i>	2	55	CN	September	40	24	ABAA
<i>Tetramorium caespitum</i>	2	50	CN	July	42	26	ABBCAB
<i>Tetramorium impurum</i>	2	60	CN	July	42	24	ABBCB
<i>Tetramorium semilaeve</i>	2	61	CN	June	40	24	ABBCAA